BLACK BOX TECHNIQUE: BOUNDARY VALUE ANALYSIS

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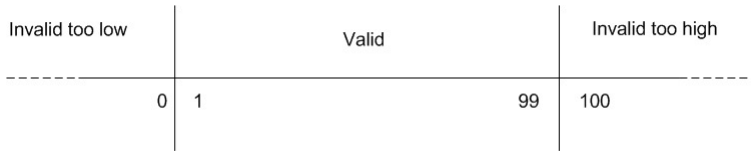
Boundary Value Analysis (BVA)

- In combination with EP:
  - Testing the borders of the equivalence class (highest and lowest value)
  - Every boundary of an equivalence class must be in a test data combination.
- BVA operates on the basis that experience shows us that errors are most likely to exist at the boundaries between partitions and in doing so incorporates a degree of negative testing into the test design
- 100% coverage of EP
- As with EP, it can be used for all Test levels
- Remember the 'circle' rule – INSIDE/OUTSIDE

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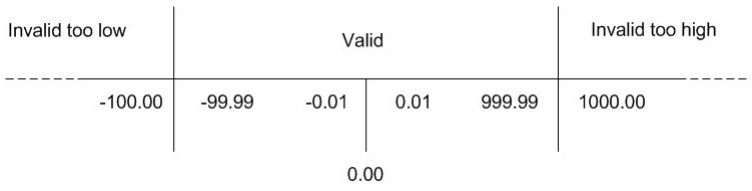
BVA Example: Integer

- "How many items would you like to order?"



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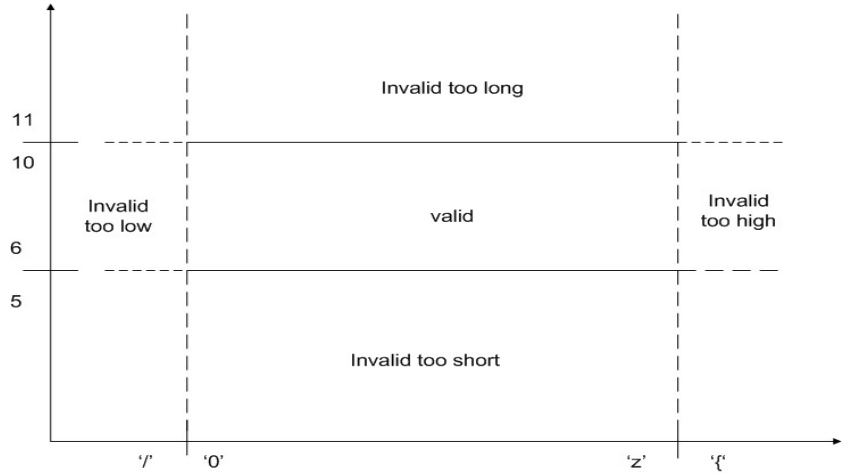
BVA Example: Real Number



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BVA Example: Character and String

- Password (6-10 character alphanumeric)

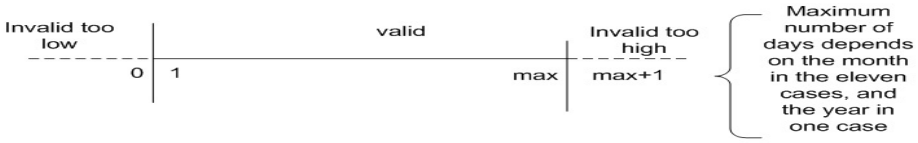


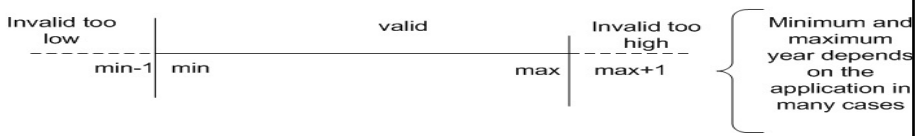
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BVA Example: Date

- Enter the departure date for your flight (MM/DD/YY)





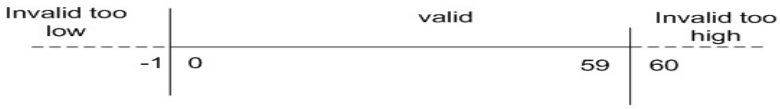


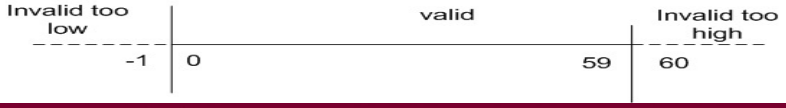
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BVA Example: Time

- Enter the departure time for your flight: (HH:MM:SS)



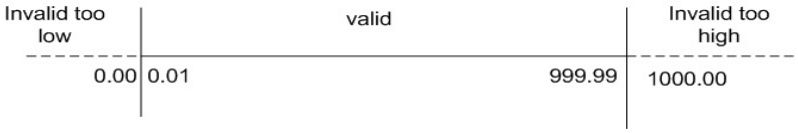




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BVA Example: Currency

- Enter a bid price (under $1000)



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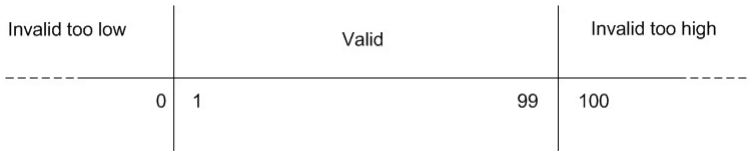
Variants of BVA

- 2-points BVA / 2-values BVA
- 3-points BVA / 3-values BVA

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2-points BVA

- In a 2-points version, for each identified equivalence class, the test values are:
  - The boundary values of this class (i.e. **min** and **max**)
  - The element just below the minimal one (i.e. **min -1**)
  - The element just above the maximal one (i.e. **max +1**)
  - Example:

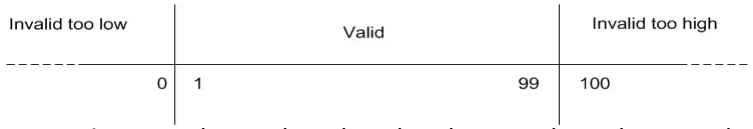


  - BV values = 0 (min-1) , 1 (min), 99 (max), 100 (max +1)

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3-points BVA

- In a 3-points version, for each identified equivalence class, the test values are:
  - The boundary values of this class (i.e., **min** and **max**)
  - The elements just below and just above the minimal one (i.e. **min - 1** AND **min + 1**)
  - The elements just below and just above the maximal one (i.e. **max -1** AND **max + 1**)



  - BV values = 0 (min-1) , 1 (min), 2 (min + 1), 98 (max – 1), 99 (max), 100 (max +1)

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BVA: Exit Criteria

- In analogy to the exit criteria of the equivalence partitioning a desired coverage of boundary values (BV) can be defined:

BV-coverage = (Amount of tested BV / Total BV) * 100 %

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